Bill of Materials

REVA AICTE IDEA Lab Portal: AWS hosting, security, payments and development tools

Ref: IDEALAB-BOM-2026-01
Date: 29 September 2026
Currency: Indian Rupees (INR)

AWS hosting + security, per month (incl. GST)	Per year (recurring)	One-time: Claude Pro (1 month)	First-year total
₹4,168	**₹50,016**	**₹2,500**	**₹52,516**

No.	Item	Specification / purpose	Provider	Quantity	Unit rate	Monthly	Annual
A. Cloud hosting: Amazon Web Services (recurring)							
1	**Compute server (EC2)**	t3.small: 2 vCPU, 2 GB RAM, Ubuntu 24.04; runs app + MongoDB in Docker	AWS, Mumbai (ap-south-1)	730 h/month	$0.0224 / hour	₹1,390	₹16,680
2	**Disk storage (EBS)**	gp3 SSD, 30 GB: database, uploaded files, local backups	AWS	30 GB	$0.0912 / GB-month	₹233	₹2,796
3	**Public IP address**	Elastic IP (static public IPv4) for the domain	AWS	730 h/month	$0.005 / hour	₹310	₹3,720
4	**Off-server backups (S3)**	Nightly database + files backup copies, ~15 GB	AWS	15 GB	$0.025 / GB-month	₹32	₹384
5	**Disk snapshots**	Weekly EBS snapshots via AWS Backup, ~10 GB changed data	AWS	10 GB	$0.05 / GB-month	₹42	₹504
6	**Email (SES)**	Admin password-reset and system emails, ~1,000 / month	AWS	1,000 emails	$0.10 / 1,000	₹8	₹96
7	**Internet data out**	Website traffic to users (first 100 GB / month free)	AWS	< 100 GB	Free tier	₹0	₹0
AWS subtotal (before tax)						₹2,015	₹24,180
GST @ 18% on AWS India invoice						₹363	₹4,356
Total A: AWS hosting						**₹2,378**	**₹28,536**
B. Extra security on AWS (recommended, recurring)							
8	**Web firewall (AWS WAF)**	Blocks common attacks (SQL/NoSQL injection, XSS, bad bots, known bad IPs) and rate-limits per IP; 1 web ACL + 5 managed rule groups	AWS	1 ACL, 5 rules, ~1M req	$5 + $1/rule + $0.60/M req	₹901	₹10,812

No.	Item	Specification / purpose	Provider	Quantity	Unit rate	Monthly	Annual
9	**CDN + DDoS protection (CloudFront + Shield Standard)**	Puts the site behind AWS's edge network so WAF can run; absorbs traffic floods; hides the server's IP	AWS	< 1 TB, < 10M req	Free tier	₹0	₹0
10	**Threat detection (GuardDuty)**	Alerts on suspicious activity: hacked server, crypto-mining, unusual logins to the AWS account	AWS	1 account	~$3 / month (usage)	₹255	₹3,060
11	**Vulnerability scanning (Inspector)**	Scans the server and Docker image for known security holes (CVEs) continuously	AWS	1 server	~$1.25 / server-month	₹106	₹1,272
12	**Secrets storage (Secrets Manager)**	Keeps JWT, database, Microsoft and Razorpay secrets out of files on disk, with rotation and access logs	AWS	5 secrets	$0.40 / secret-month	₹170	₹2,040
13	**Monitoring and alarms (CloudWatch)**	Server logs off the machine, alarms for downtime/high CPU/disk full, emailed to admins	AWS	~1 GB logs, 10 alarms	~$1 / month	₹85	₹1,020
Security subtotal (before tax)						₹1,517	₹18,204
GST @ 18% on AWS India invoice						₹273	₹3,276
Total B: Extra security						**₹1,790**	**₹21,480**
RECURRING TOTAL (A + B)						**₹4,168**	**₹50,016**

No.	Item	Specification / purpose	Provider	Quantity	Unit rate	Amount
C. One-time costs						
14	**Claude Pro subscription**	AI assistant (Claude Code) used to build, secure and document the portal; one month only	Anthropic	1 seat × 1 month	₹2,500 / month	₹2,500
Total C: One-time						**₹2,500**

No.	Item	What it covers	Rate	Example
D. Payment gateway: Razorpay (usage-based, not in the totals)				
15	**Razorpay account and APIs**	Orders API, payment verification, webhooks, test mode, dashboard	Free: no setup or annual fee	₹0
16	**Domestic payments**	UPI, RuPay, Indian debit/credit cards, net banking, wallets	2% per payment + 18% GST on the fee	₹10,000 collected → ₹236
17	**International / premium cards**	Foreign cards, Amex, Diners, corporate cards	3% per payment + 18% GST on the fee	₹1,000 collected → ₹35
18	**Settlement**	Money reaches the lab's bank account (T+2 working days)	Included	₹0

No.	Item	Specification / purpose	Provider	Quantity	Unit rate	Monthly	Annual
E. Included at no additional cost							
19	**Database**	MongoDB 7 Community Edition (in Docker)	MongoDB Inc. (open source)	-	Free / existing	₹0	₹0
20	**Web server / HTTPS**	nginx + Let's Encrypt SSL certificate (auto-renew)	Open source / Let's Encrypt	-	Free / existing	₹0	₹0
21	**Runtime**	Node.js 22, Docker Engine + Compose	Open source	-	Free / existing	₹0	₹0
22	**Sign-in**	Microsoft Entra ID app registration (REVA's existing Microsoft 365)	Microsoft / REVA IT	-	Free / existing	₹0	₹0
23	**Domain**	Subdomain, e.g. idealab.reva.edu.in (REVA DNS)	REVA IT	-	Free / existing	₹0	₹0
24	**Source code + CI**	GitHub repository and GitHub Actions (tests, Docker build)	GitHub (free tier)	-	Free / existing	₹0	₹0

Cost summary

Summary	
A. AWS hosting (monthly, incl. GST)	₹2,378
B. Extra security (monthly, incl. GST)	₹1,790
Recurring per month (A + B)	**₹4,168**
Recurring per year (A + B) × 12	**₹50,016**
C. One-time: Claude Pro, 1 month	₹2,500
FIRST-YEAR TOTAL (A + B) × 12 + C	**₹52,516**

Without the optional security (B): first year = ₹31,036

Notes and assumptions

1. AWS prices are on-demand list rates for the Mumbai region (ap-south-1), converted at an assumed **1 USD = ₹85**; AWS India invoices add 18% GST. Usage-based items (WAF requests, GuardDuty, CloudWatch) are estimates for a college portal. Confirm with the AWS Pricing Calculator (calculator.aws) before purchase.
2. Claude Pro is budgeted for **one month only** (₹2,500 as provided). Renew only if more development work is needed.
3. Section B is recommended, not required: the portal already has its own protections (strict security headers, rate limits, injection checks, append-only sign-in log). B adds a layer in front of the server. Setting up CloudFront + WAF, Secrets Manager and the alarms is a configuration task that is not yet in the repository.
4. Razorpay has no fixed fee; the lab pays only a percentage of what students pay. Rates are Razorpay's standard plan; check the final rates on the lab's Razorpay account. Fees can instead be passed on to students as a convenience fee.
5. Server size fits a college portal with a few hundred to a few thousand users. Upgrading to t3.medium (4 GB RAM) adds about ₹1,400 / month before GST. A 1-year EC2 Savings Plan lowers the server cost by roughly 30–40%.
6. Microsoft sign-in (with REVA's MFA policies) uses REVA's existing Microsoft 365 at no cost. The domain is assumed to be a REVA subdomain; a separate domain would cost about ₹800–1,500 / year. Not included: staff time and lab hardware.

Prepared by	Reviewed by	Approved by
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